

NCDs

NON-COMMUNICABLE DISEASES

PREDICTION

This aims to predict the risk of developing non-communicable diseases such as diabetes among older adults by utilizing personal information, lifestyle behaviors, and health examination data.





CONCEPTUAL DESIGN



Conceptual Design

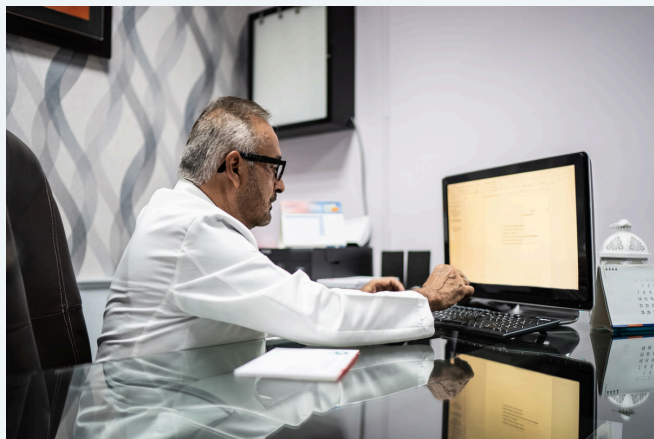
MOVING TARGET



Conceptual Design

USE CASE DIAGRAM

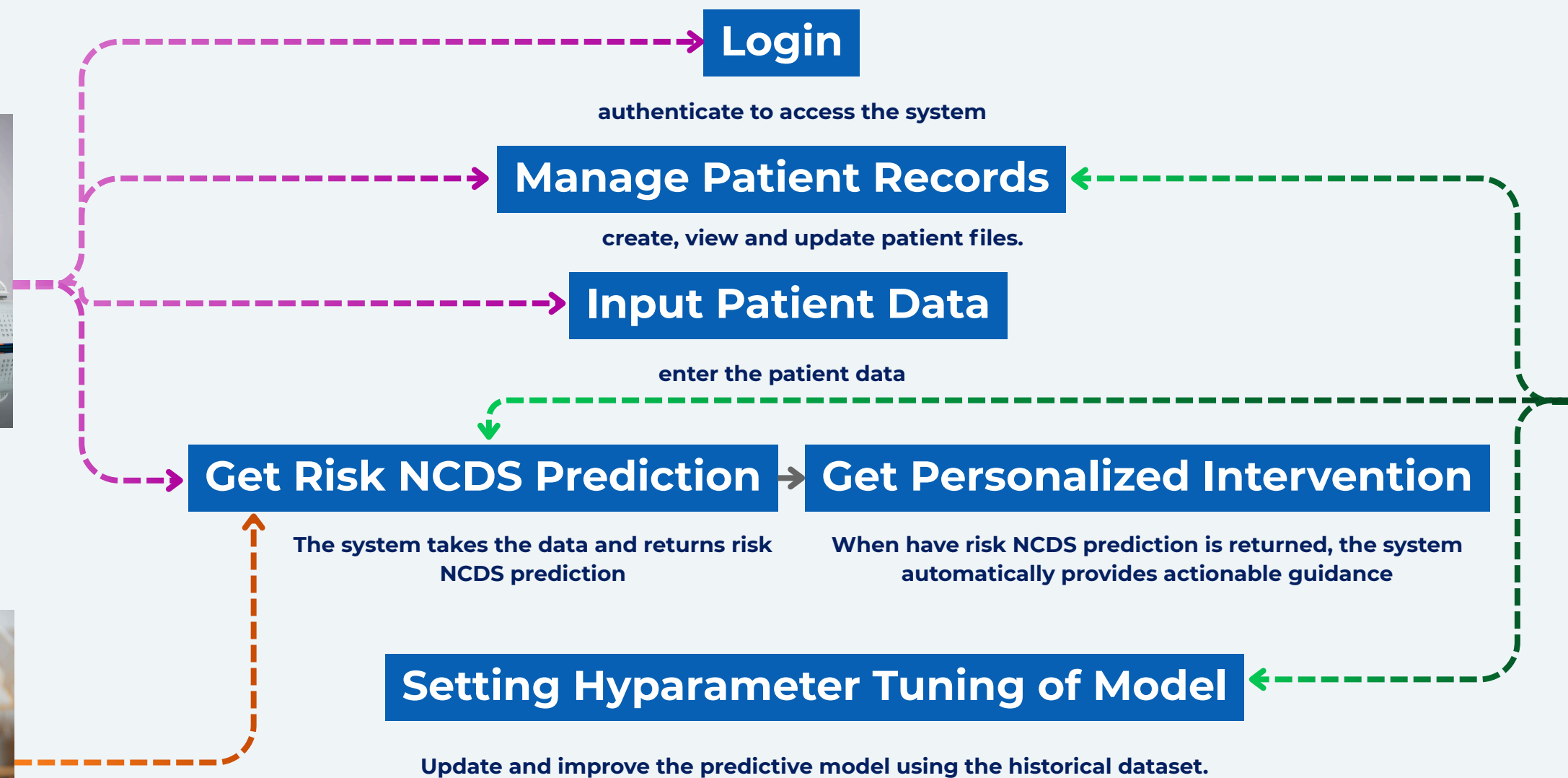
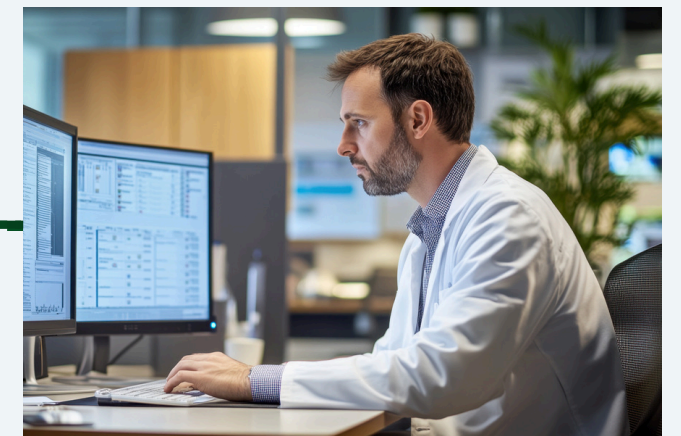
Related hospital personnel



Patients

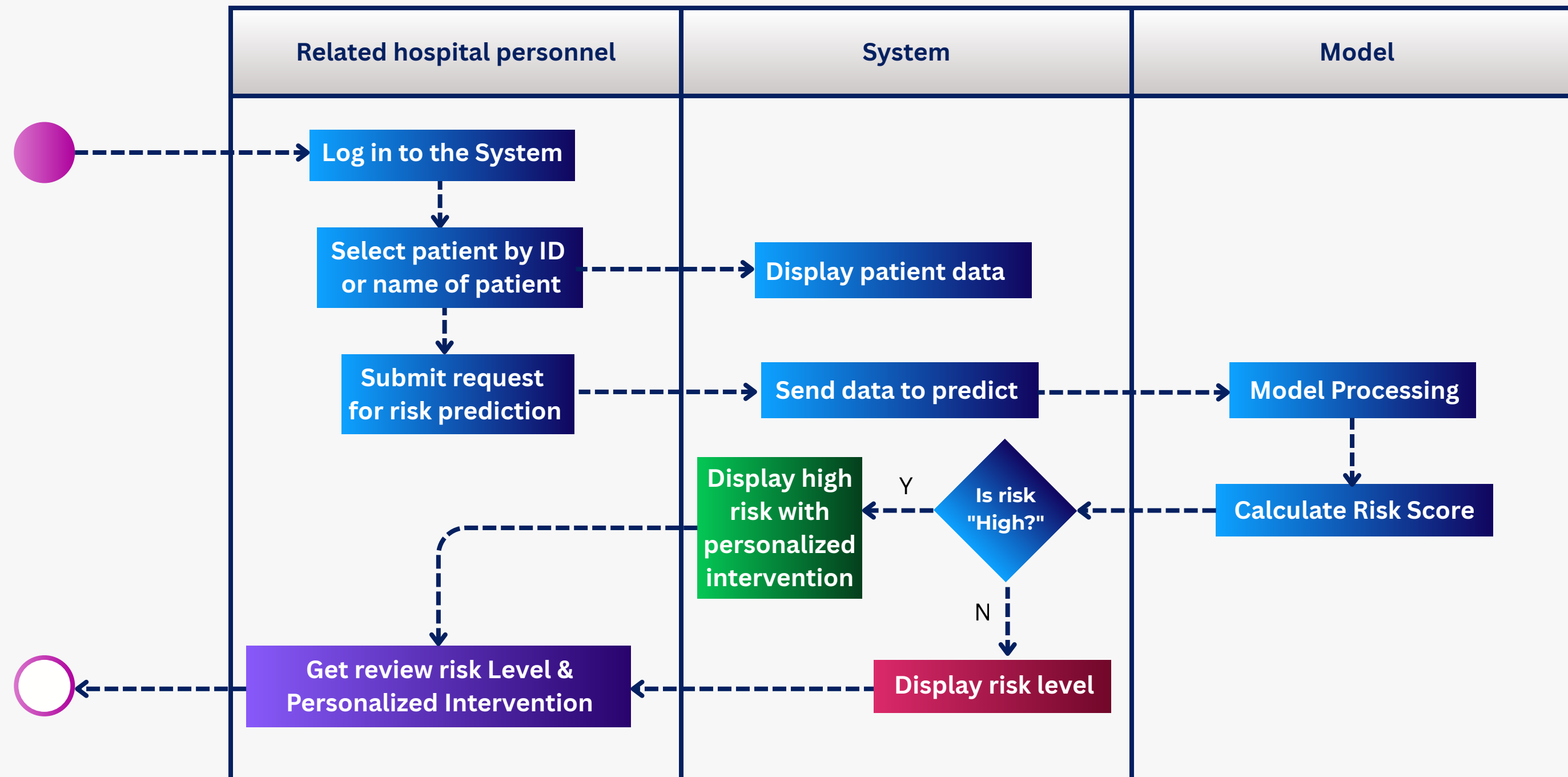


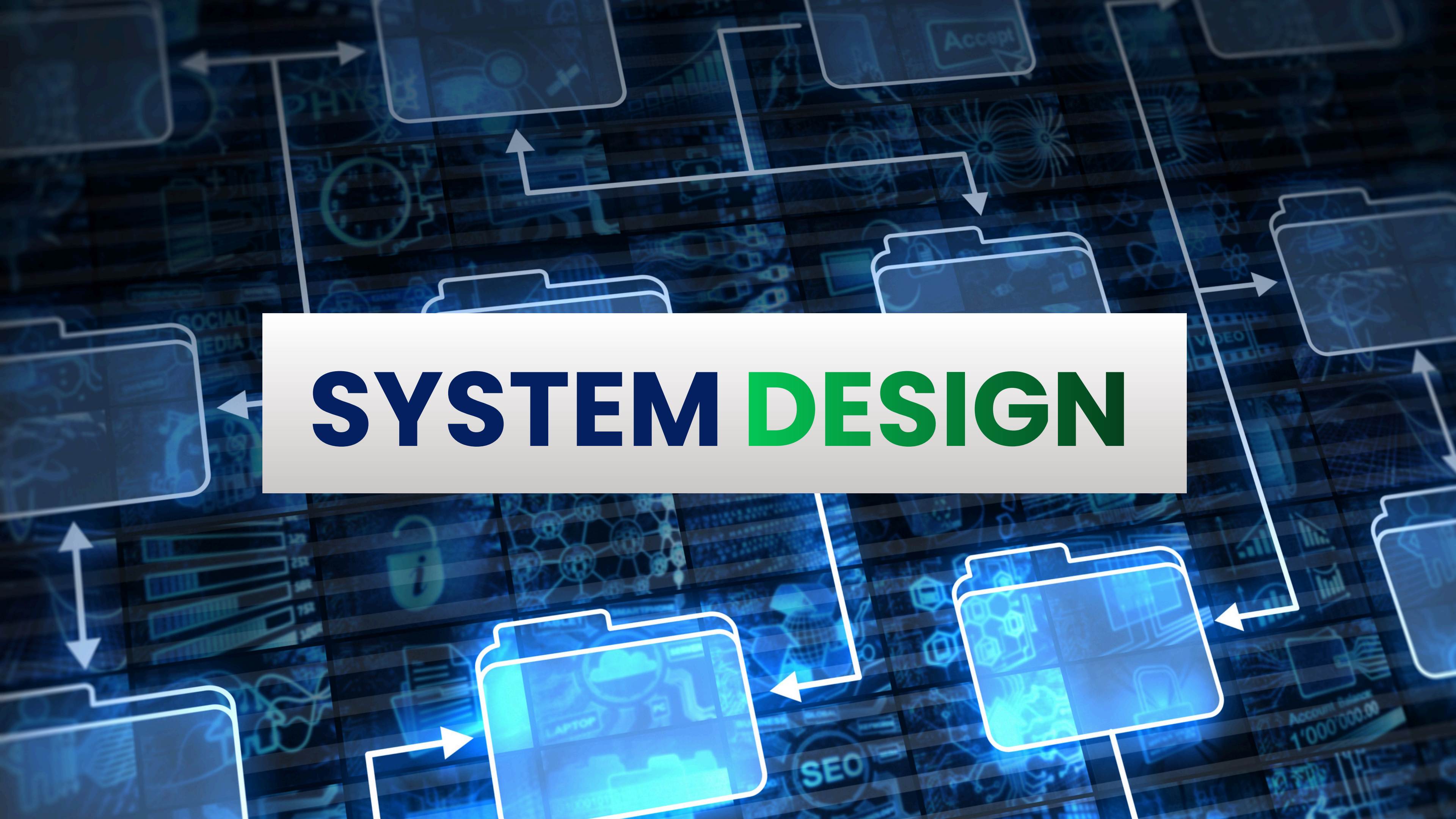
Data Scientist



Conceptual Design

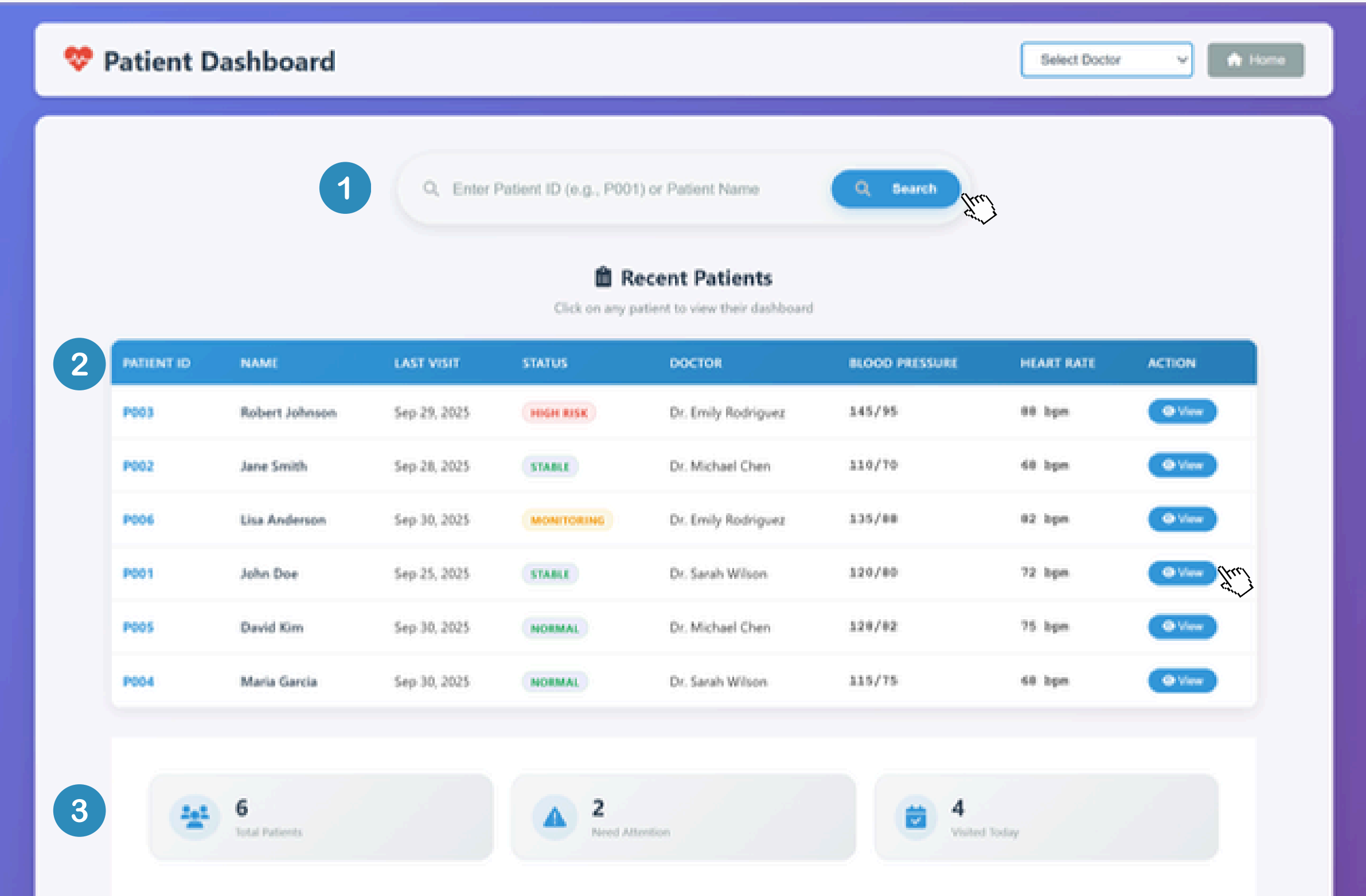
ACTIVITY DIAGRAM





SYSTEM DESIGN

UI FUNCTIONS & SYSTEM SEQUENCE

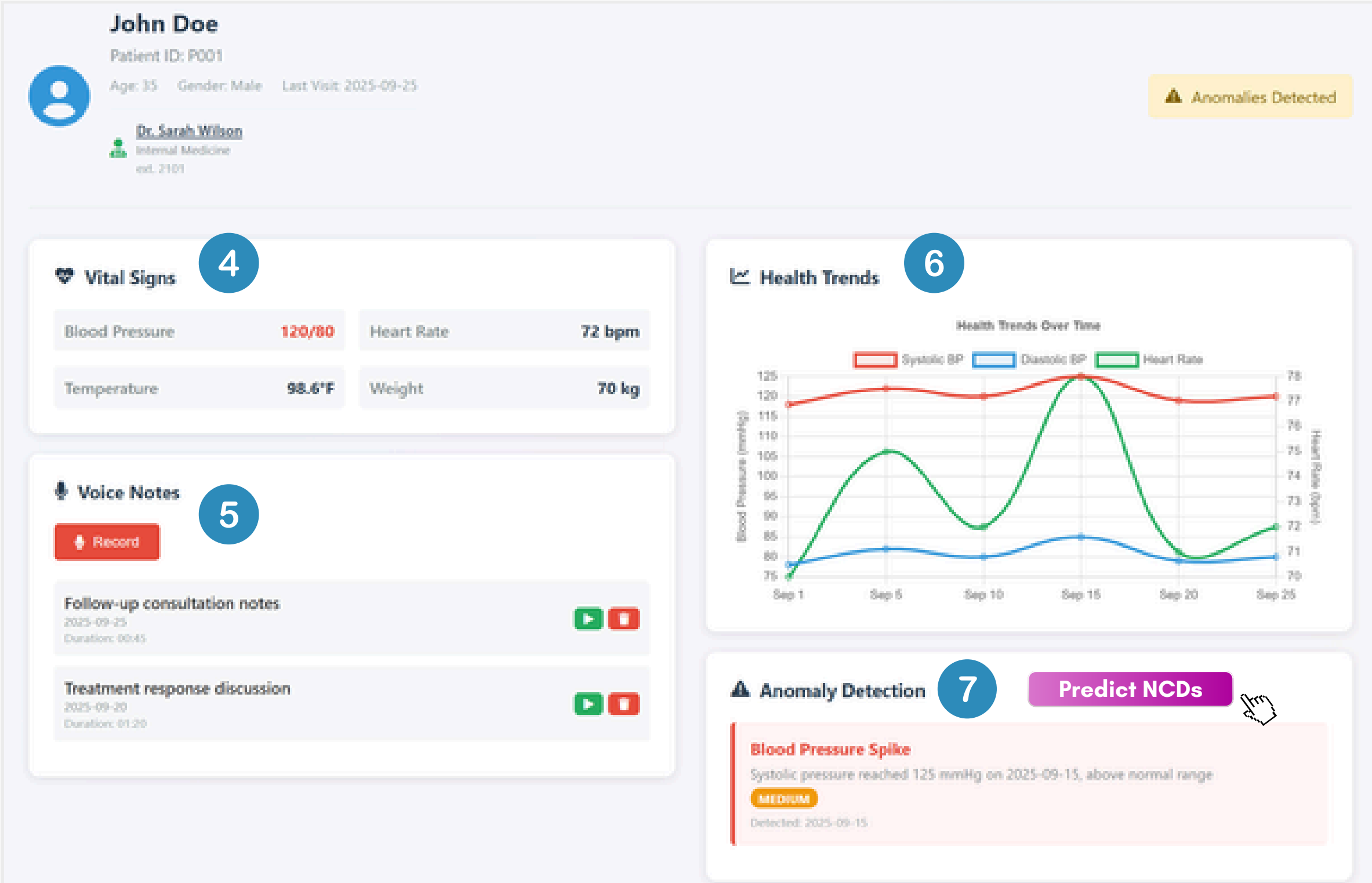


- 1 Patient Search**
by patient ID or name of patient
- 2 Patient Table**
show last visit, status, doctor and information of health
- 3 Patient Dashboard**
show total patient, NCDs attention and amount visited today



Specification Design

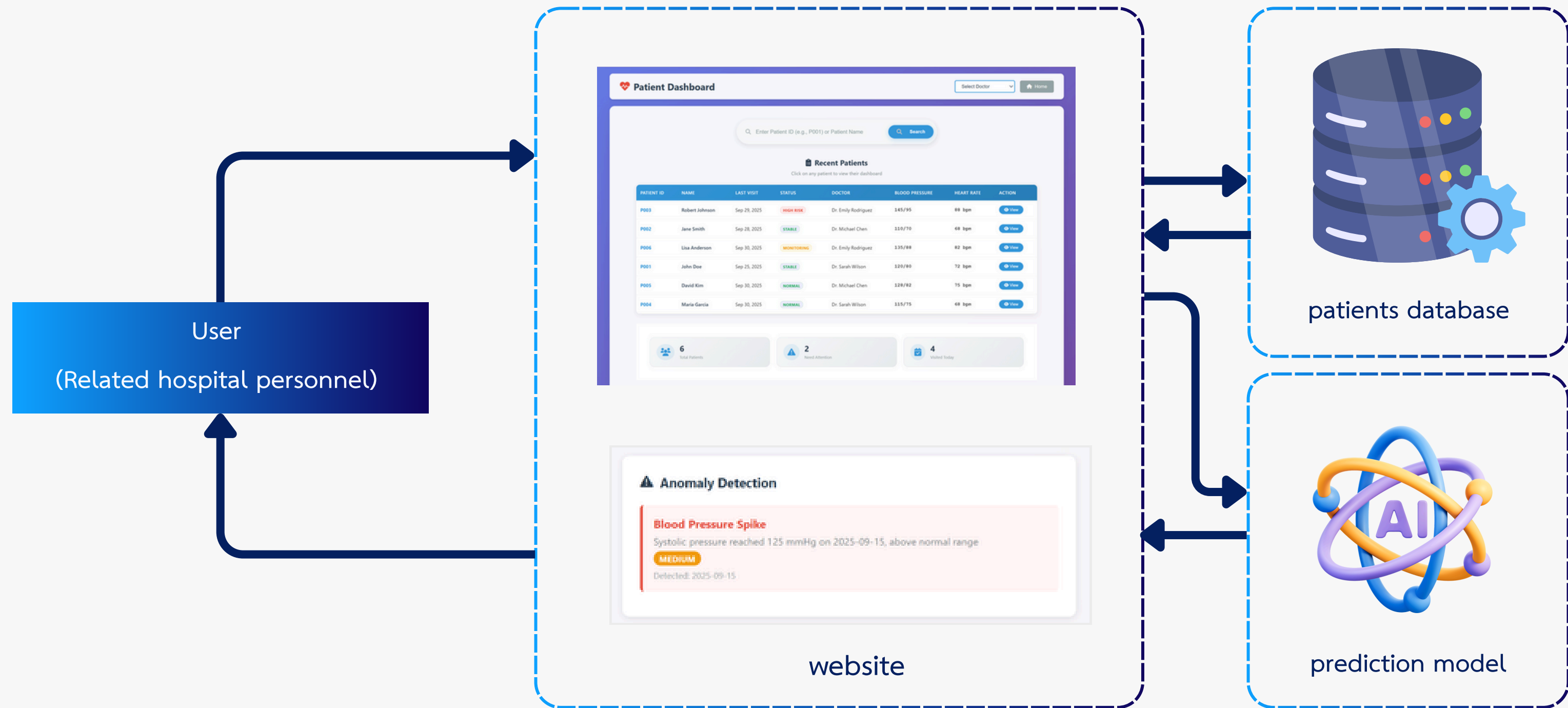
UI FUNCTIONS & SYSTEM SEQUENCE



- 4 Vital Signs**
show blood pressure, heart rate, temperature and weight
- 5 Voice Notes Record**
record result of treatment by doctor's voice
- 6 Health Trends**
show heath trends such as blood pressure
- 7 Anomaly Detection**
show risk of NCDs

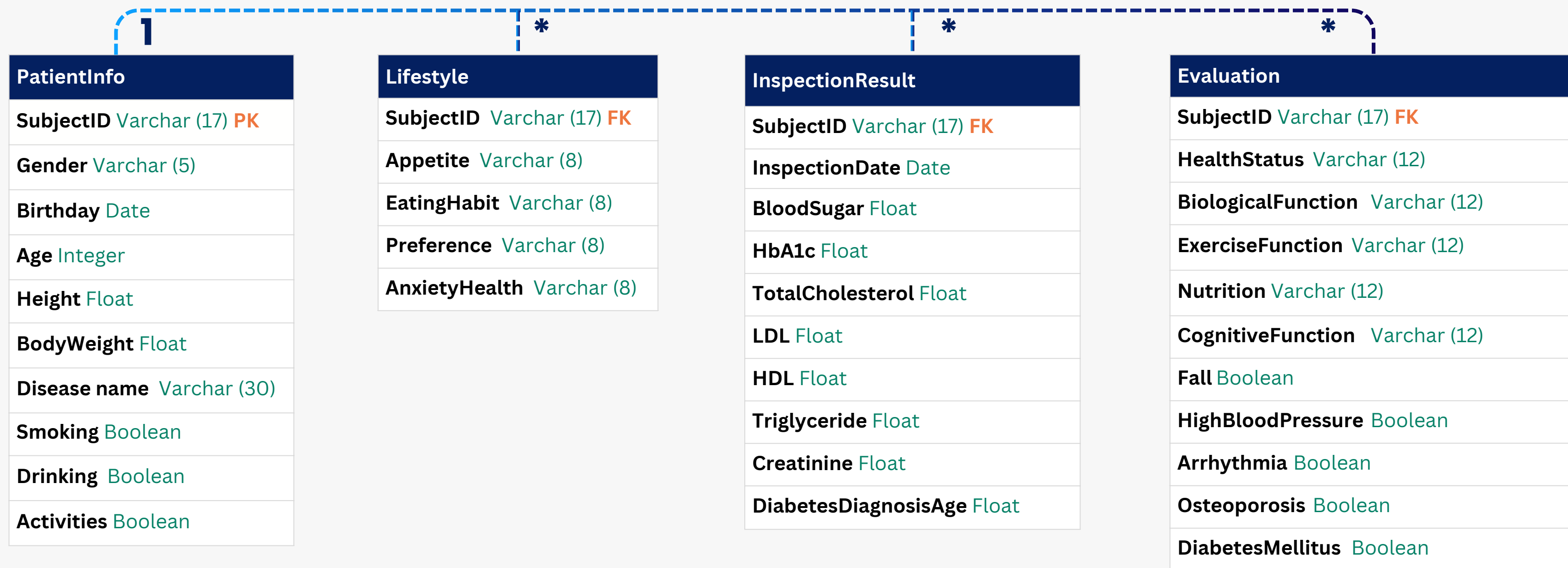
Specification Design

SYSTEM CONFIGURATION



Conceptual Design

DATABASE DIAGRAM



Master table

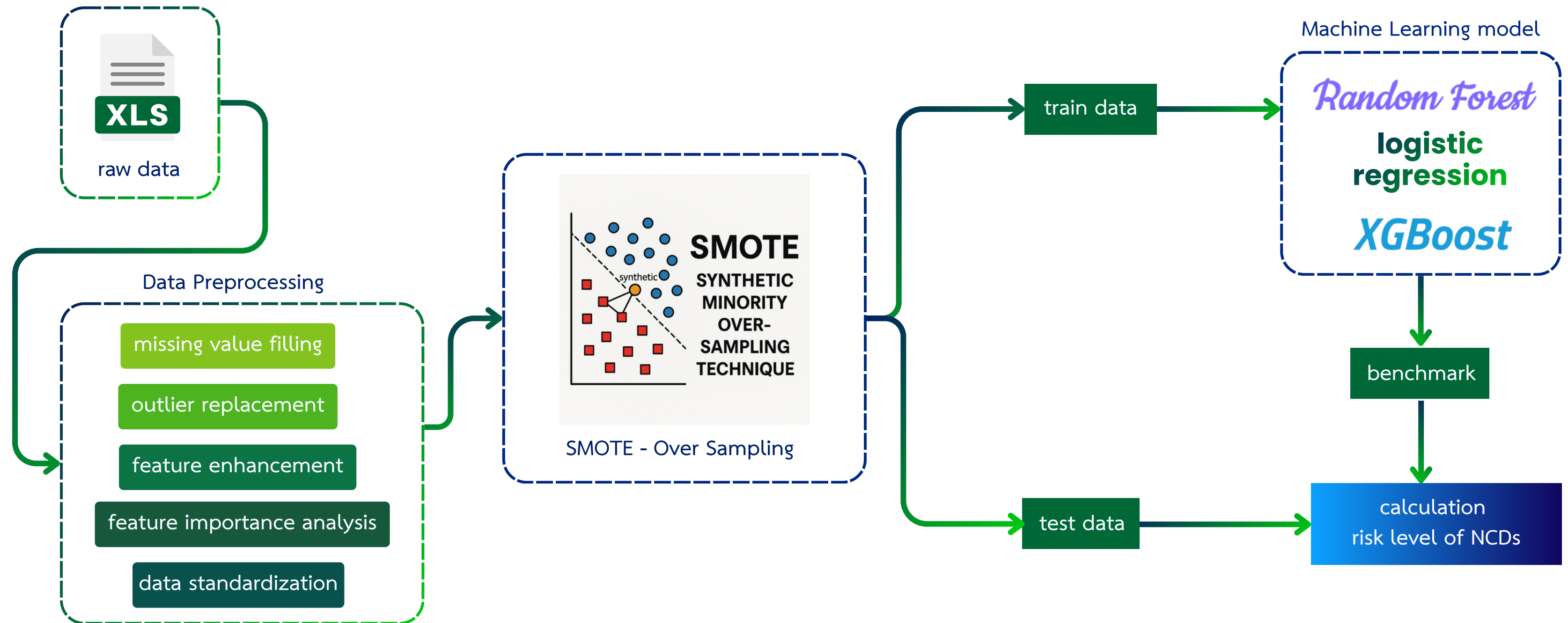


OPERATING ENVIRONMENT

Component	Specification
Operating System	Windows 11 Pro, 64-bit or Linux Ubuntu 22.04 LTS
Database Management System (DBMS)	MySQL 8.0 / PostgreSQL 15 / SQL Server 2022
Hardware Requirement	CPU 4-core, RAM 8GB, Storage 100GB
Application Language / Framework	Python 3.11, Django 4.2 / Java 17, Spring Boot 3.2
Browser / Client	Chrome 120+, Edge 120+

Structural Design

METHODOLOGY



Structural Design

PYTHON LIBRARY



Data manipulation & analysis



Numerical computing with arrays



Machine learning & model evaluation



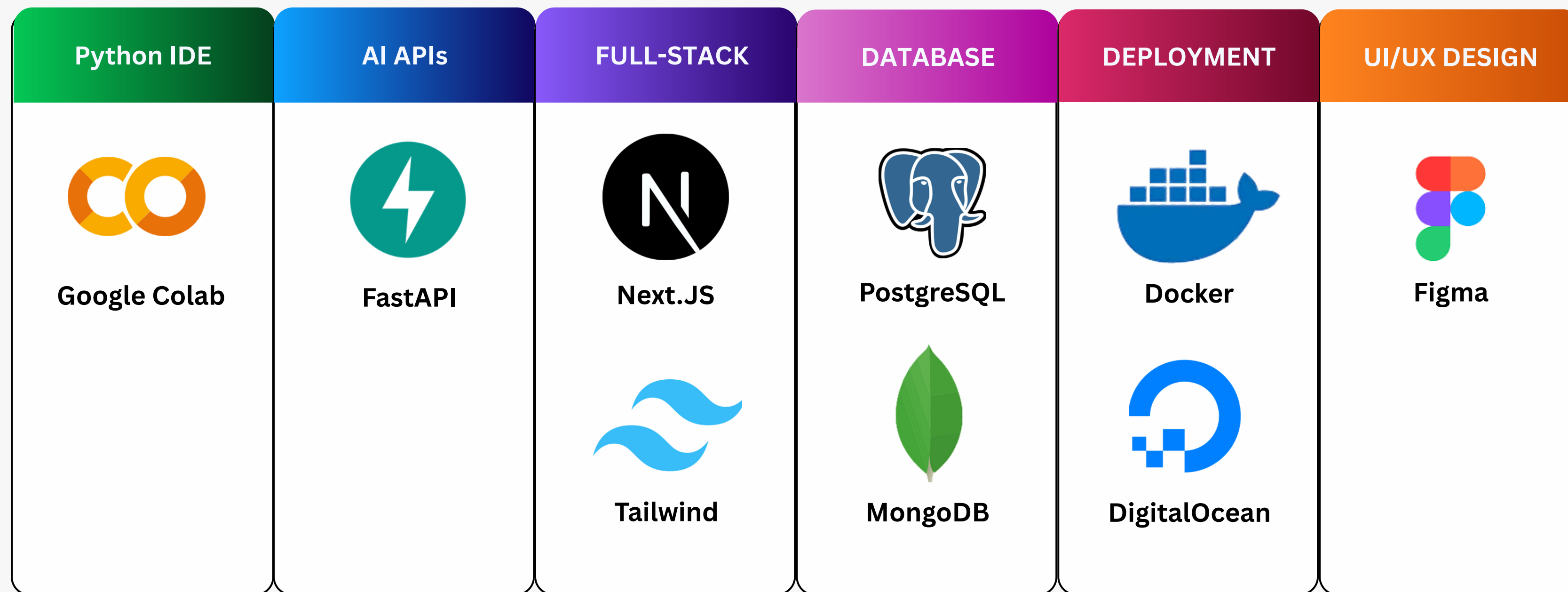
Data visualization & plotting

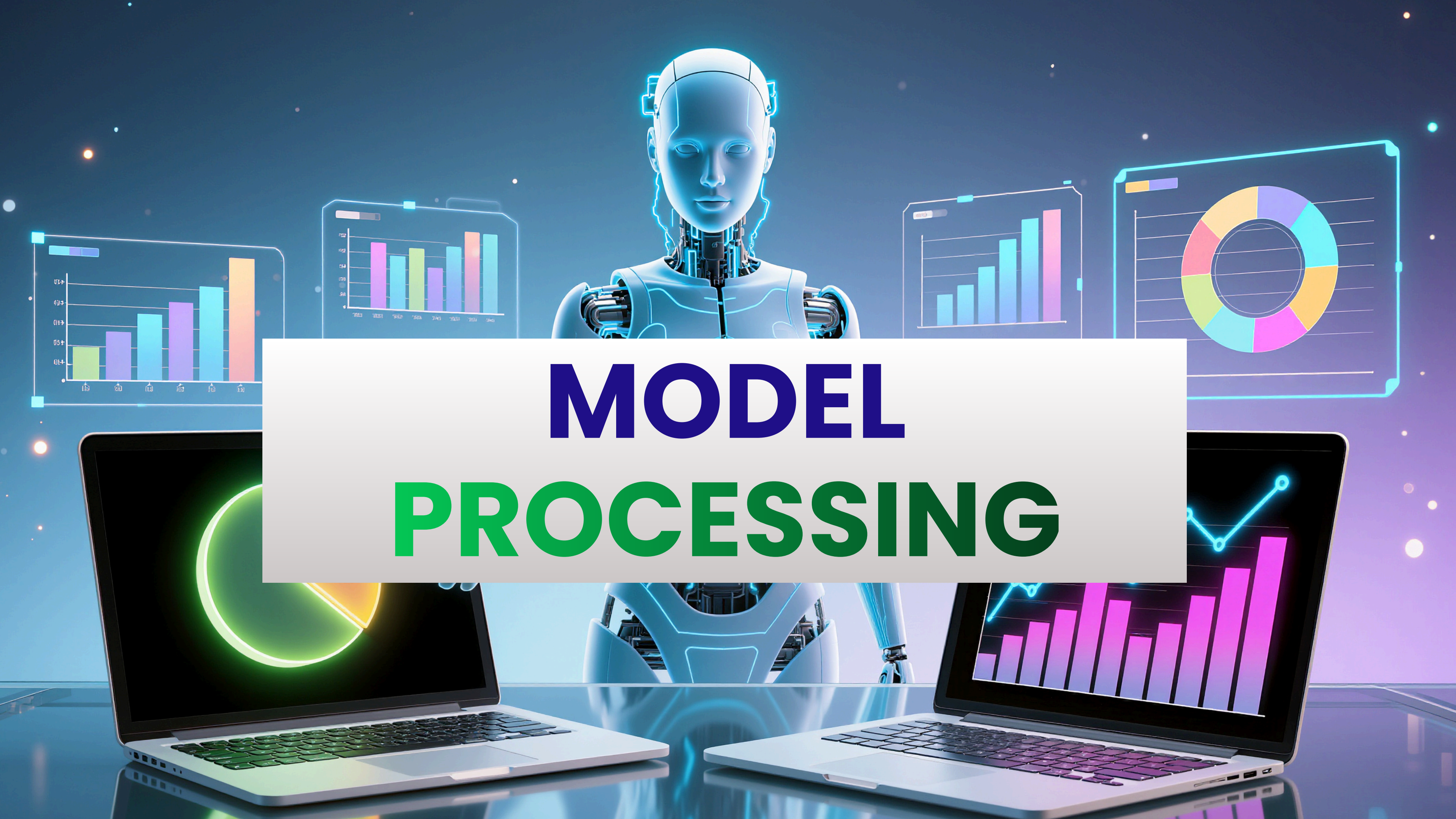


Stylish data visualization

Structural Design

TECH STACK

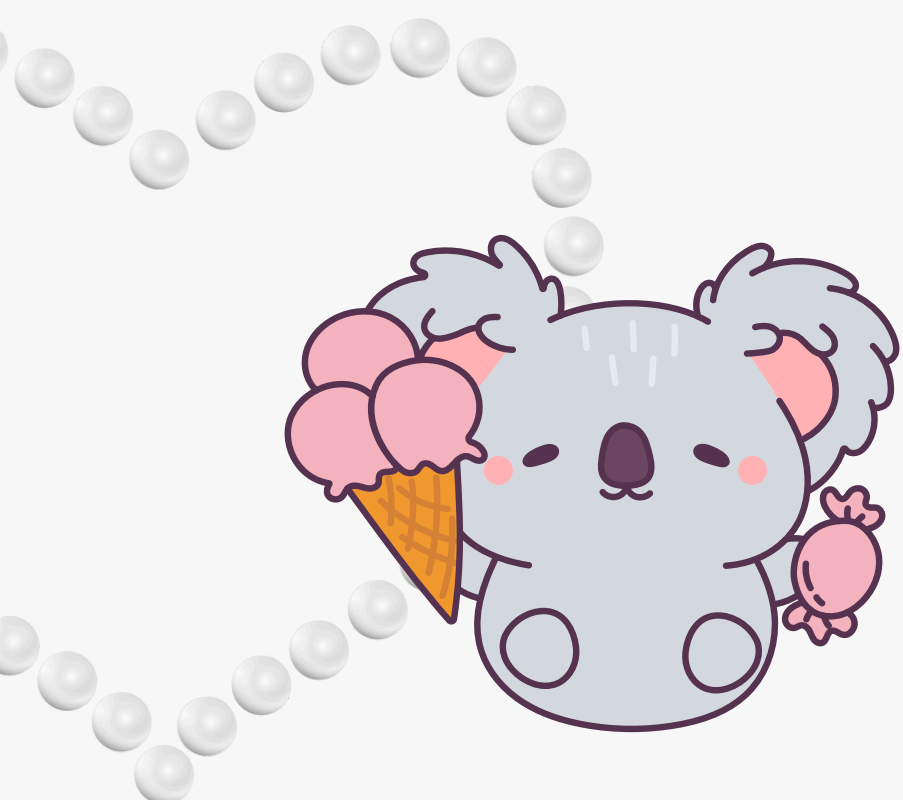
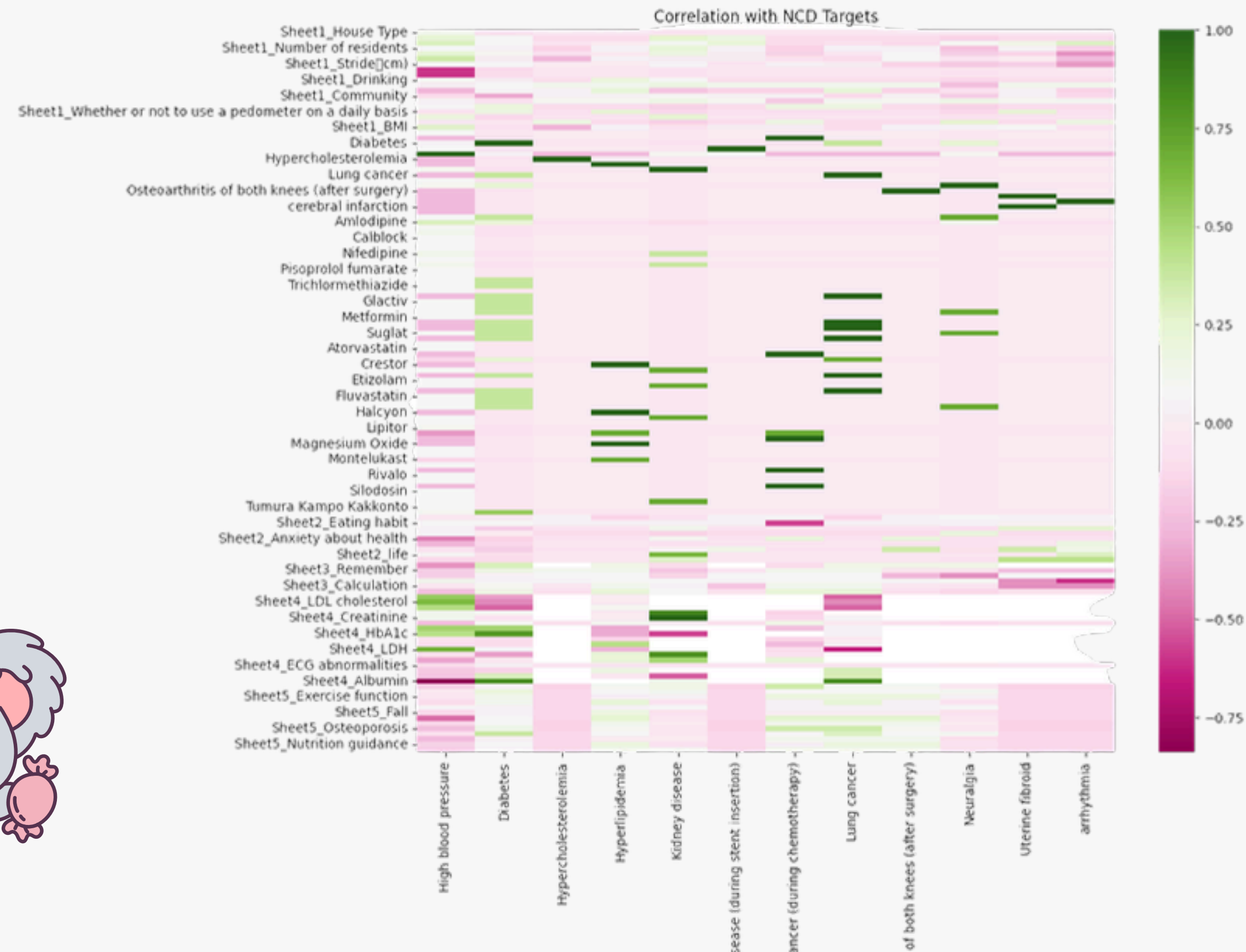


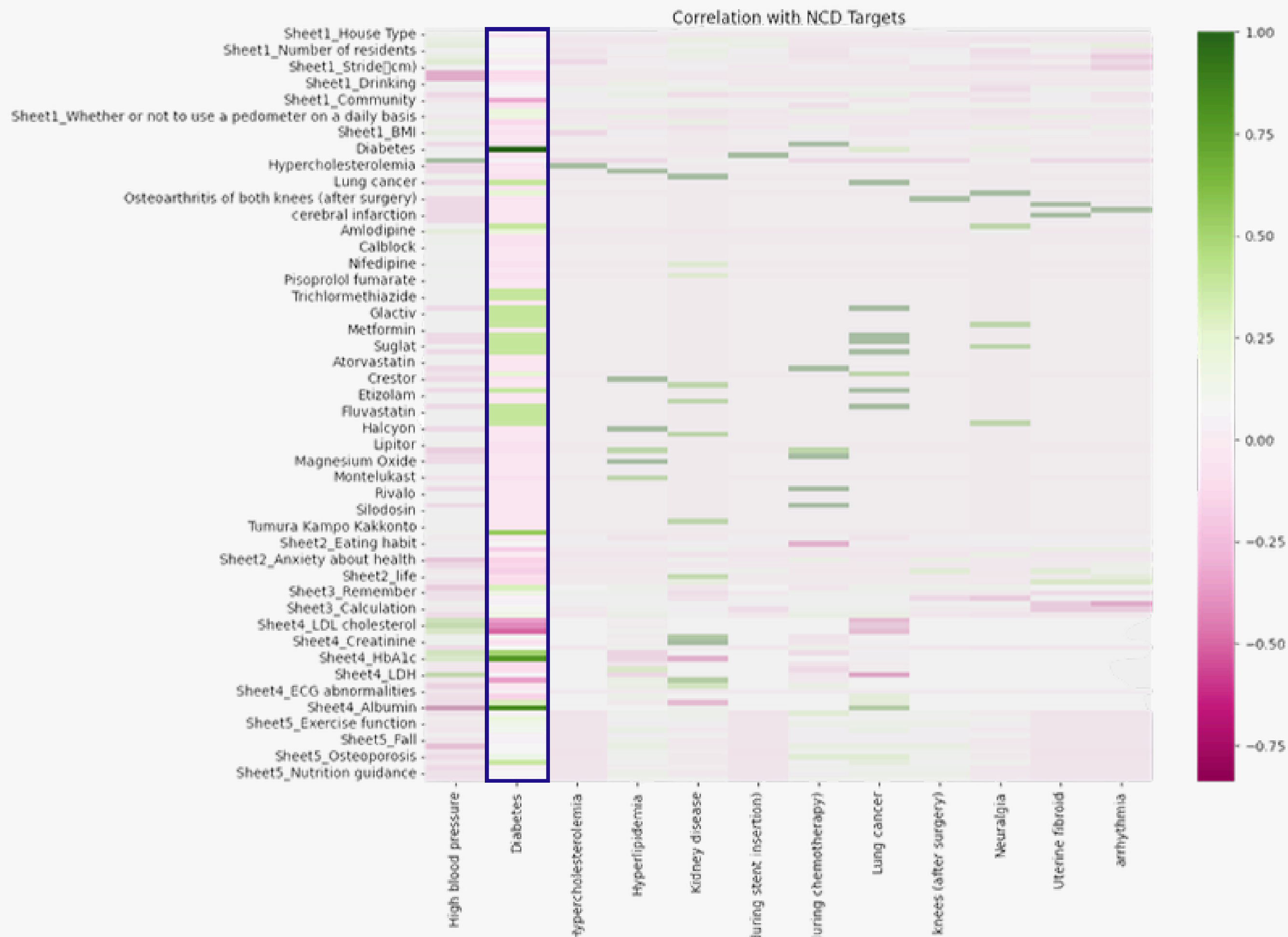


MODEL PROCESSING

Model Processing

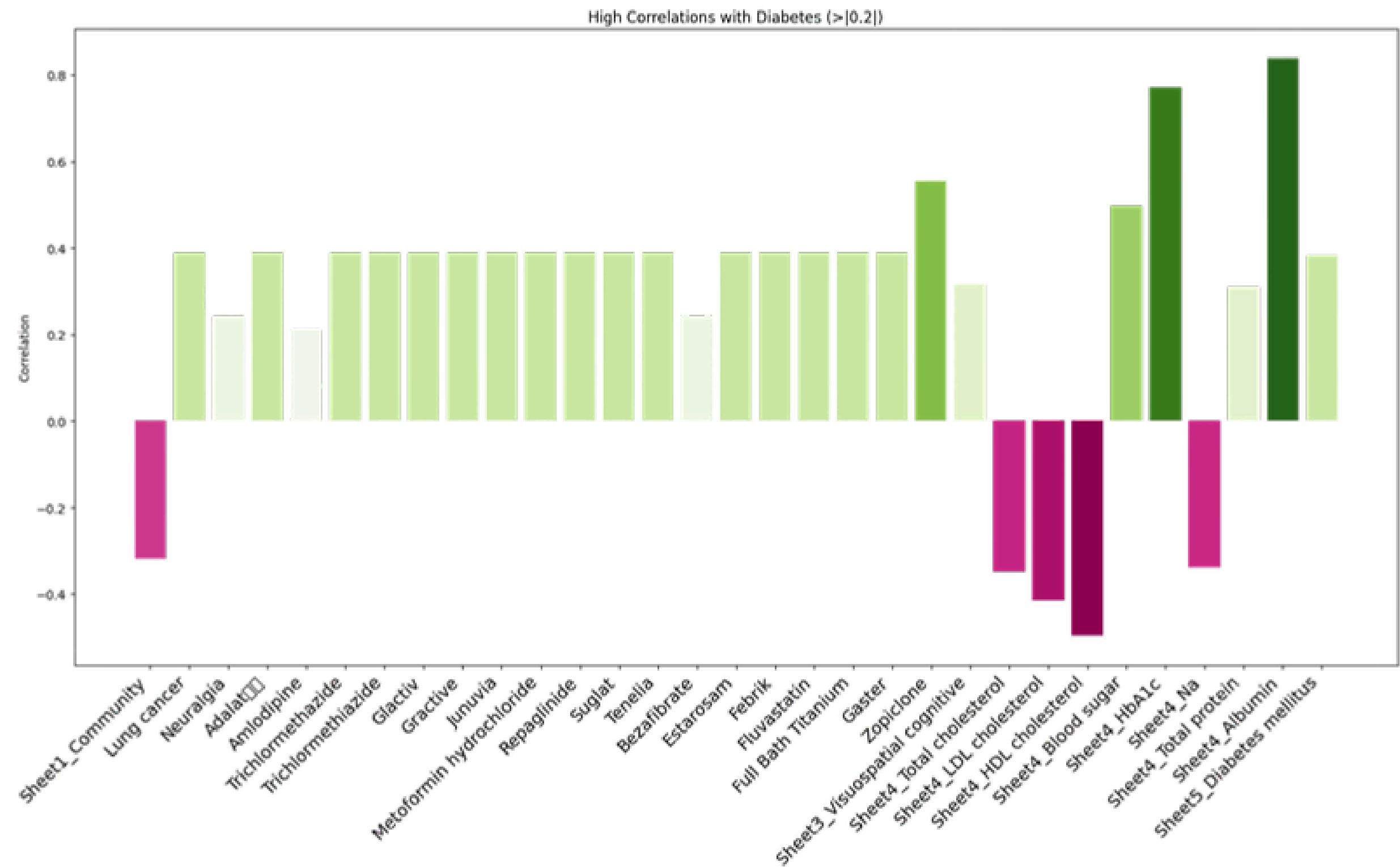
EDA





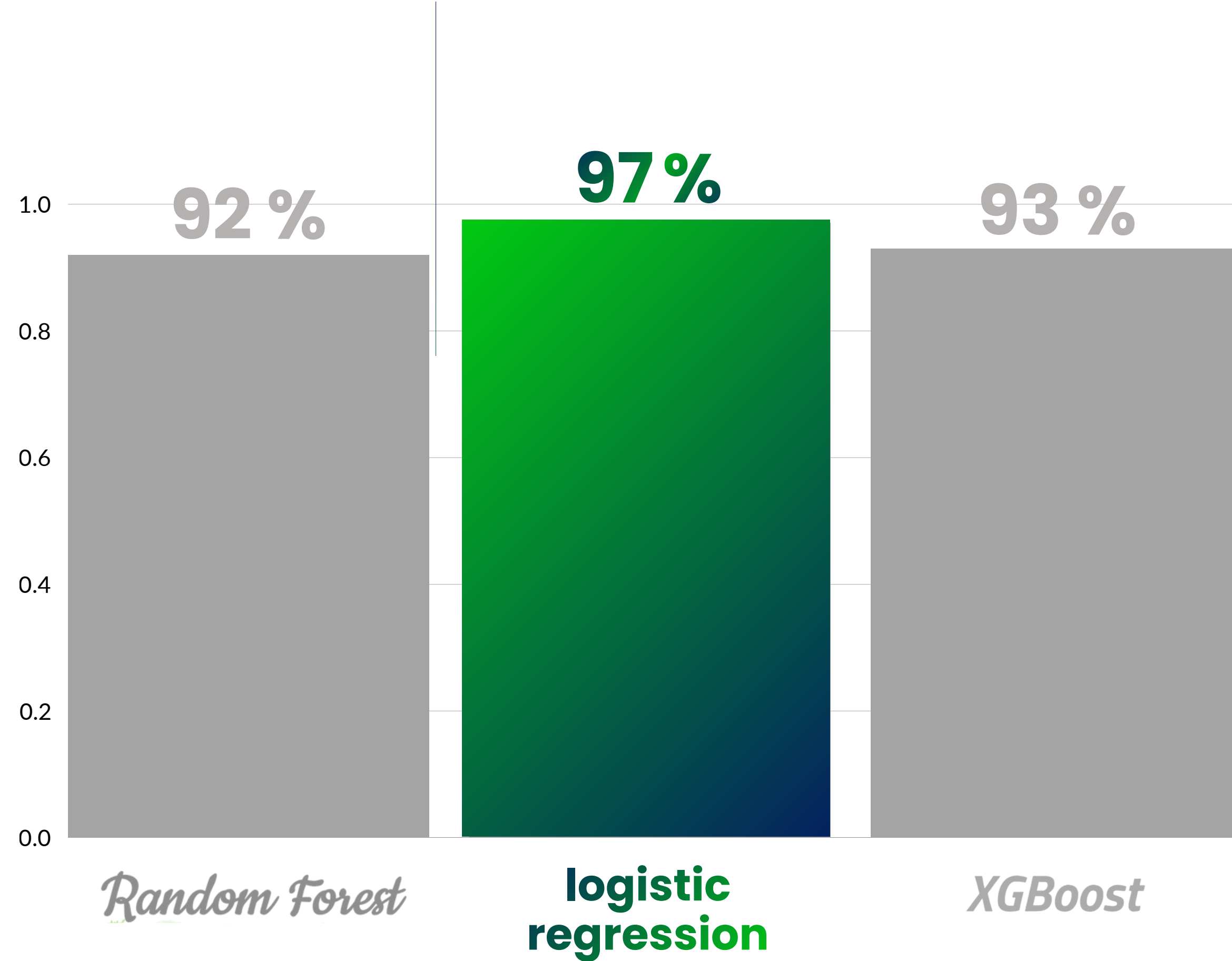
Model Processing

EDA



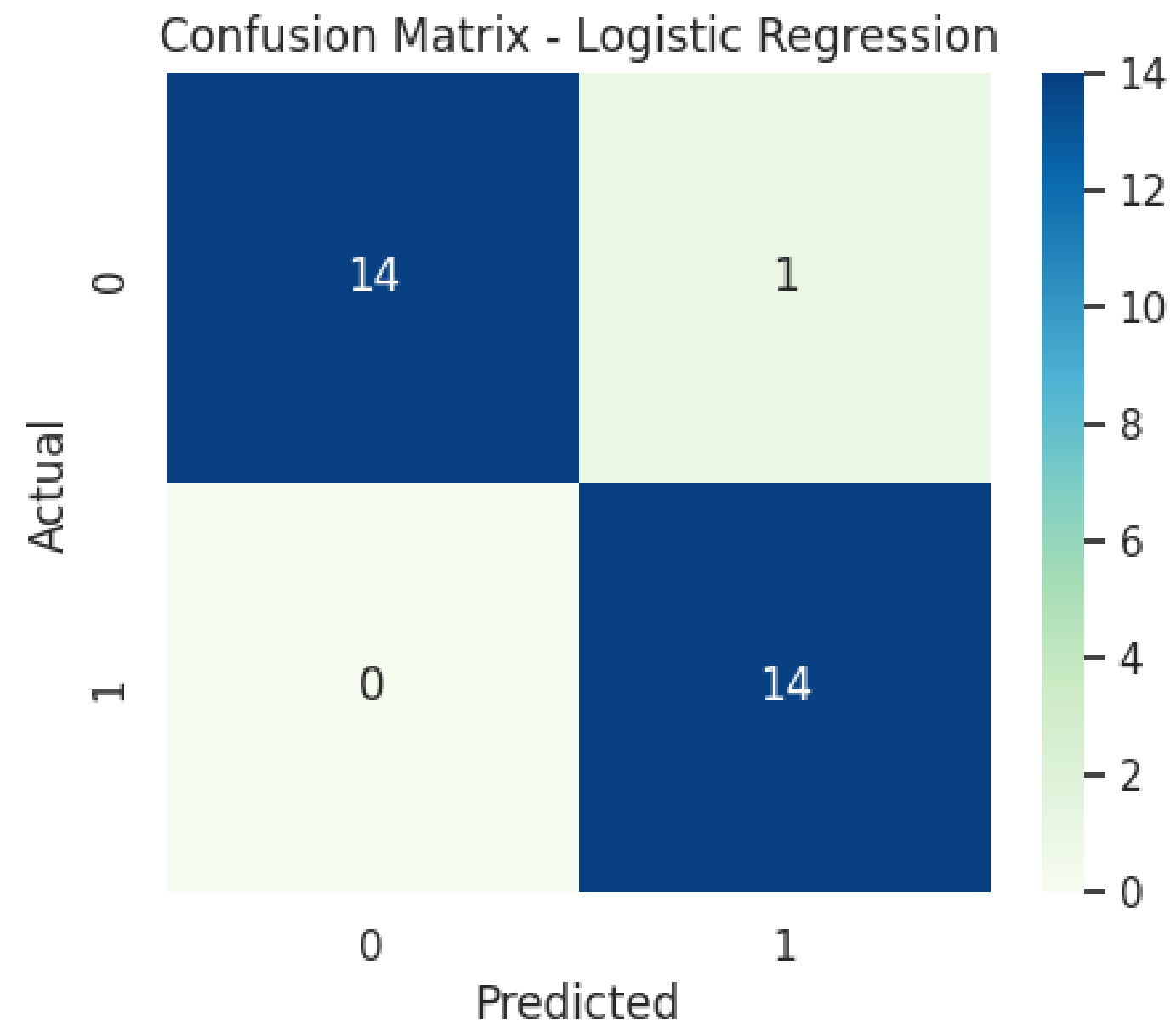
Model Processing

MODEL



Model Processing

MODEL



Accuracy

97 %

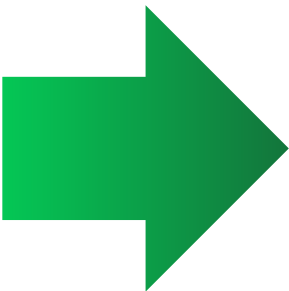
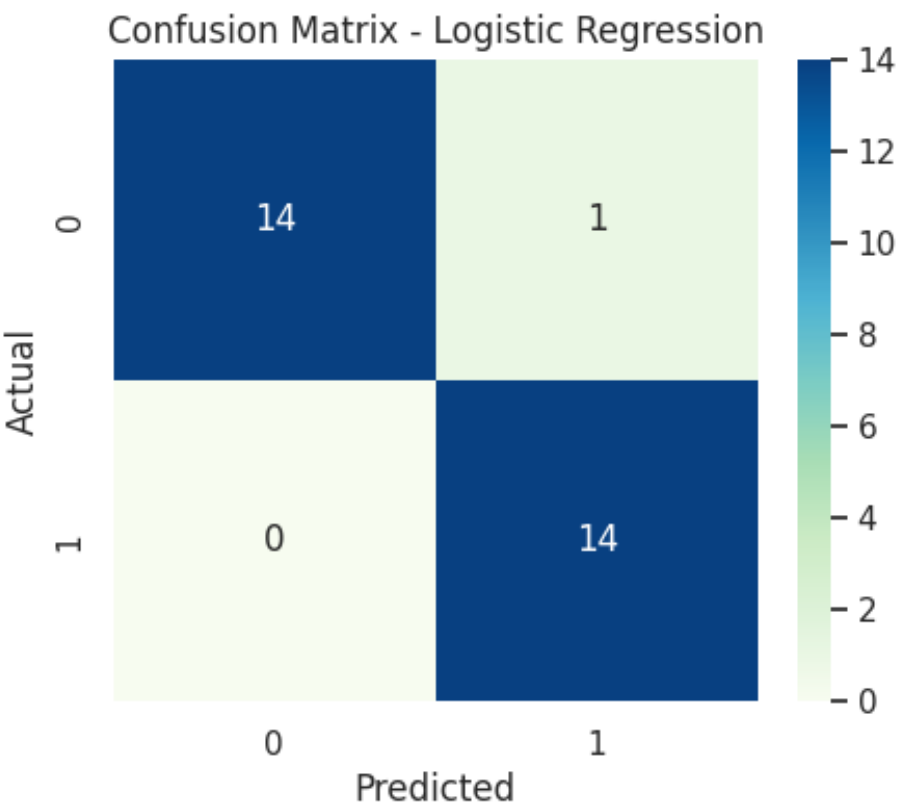
classification	precision	recall	f1-score
No diabetes	1.00	0.93	0.97
Has diabetes	0.93	1.00	0.97

Model Processing

MODEL



28 EXAMPLES



Probability (%)	Risk Level
6.832433047088343	Low Risk
0.10560155971731891	Low Risk
96.69672996294202	High Risk
96.11832544016843	High Risk
11.46982037825066	Low Risk
94.19146821907687	High Risk
0.6902525295412796	Low Risk
1.7628915817386137	Low Risk
60.417125817382946	Medium Risk
98.94115351479951	High Risk
7.475477895893771	Low Risk
84.43340580614482	High Risk
96.26596141527172	High Risk
2.835318298724958	Low Risk
95.04376362210995	High Risk
96.3541913902479	High Risk
0.24411306336699226	Low Risk
18.082641645375297	Low Risk
4.476662088987894	Low Risk
99.33108264504352	High Risk
0.836487723127201	Low Risk
1.2158931934283352	Low Risk
95.26557233405363	High Risk
2.393070115023066	Low Risk
99.7564525406468	High Risk
96.48448830964765	High Risk
35.89753583188518	Medium Risk
82.2236887480989	High Risk
60.141802845174674	Medium Risk

- 0.00 – 0.29

Low Risk
- 0.30 – 0.69

Medium Risk
- 0.70 – 1.00

High Risk

The background of the image shows a person's profile on the left, looking towards a laptop on the right. The laptop screen displays a cityscape. Overlaid on the image are several white circular icons: a group of three people in the top left, a globe in the top center, a gear with 'QC' in the bottom left, a thumbs up in the bottom center, a clipboard with a checkmark in the bottom left, and a clipboard with a pencil in the bottom center. A large, stylized gear icon is also visible in the upper right quadrant. The text 'UNIT TEST & OPERATION TEST' is centered in a white rectangular box.

UNIT TEST & OPERATION TEST

Unit Test & Operation Test

UNIT TEST

Activity Level

Stay room → Low

Join / Hobby → Medium

Hard work → High

input_activity	activity_level_output
Work (5 days / week)	High
Housing complex manager	High
teacher of Karaoke	Medium
Handmade class (every day)	Medium
Stay in the room all day	Low
0	Low

Diabetes Risk Level

0.00 – 0.29 → Low

0.30 – 0.69 → Medium

0.70 – 1.00 → High

prob	risk_level_output
0	Low Risk
0.1	Low Risk
0.29	Low Risk
0.3	Medium Risk
0.55	Medium Risk
0.69	Medium Risk
0.7	High Risk
0.95	High Risk

BMI Level

< 18.5 → Underweight

18.5 – 24.9 → Normal

25 – 29.9 → Overweight

30 + → Obese

weight	BMI_Level
17.0	Underweight
18.4	Underweight
18.5	Normal
23.0	Normal
24.9	Normal
25	Overweight
28.5	Overweight
29.9	Overweight
30	Obese
32	Obese

Unit Test & Operation Test

OPERATION TEST

Test ID	Test Scenario	Test Steps	Expected Result	Status
OT01	Open dashboard page	1. Log in as doctor 2. Navigate to dashboard	Dashboard loads successfully showing patient list, selection dropdown, and latest data	T
OT02	Select a doctor	1. Select a doctor from “Select Doctor” dropdown 2. Verify UI updates	Patients list updates to show that doctor’s patients	T
OT03	View patient details	1. Click on a patient record 2. Display patient info and vitals	Patient info page shows age, gender, last visit, vitals, etc.	T
OT04	Verify data display	Browse patient list	All fields (ID, BP, HR, Temp, Weight) are visible and correctly formatted	T
OT05	Search for a patient	1. Enter partial/full name in search box 2. Execute search	List filters to matching patient(s)	T
OT06	Navigation consistency	Use breadcrumb/back buttons to return to list	Page returns without losing context	T

The background of the slide is a collage of business-related images. The top half shows the upper bodies of several people in professional attire, including dark blue and white blouses. The bottom half features a close-up of multiple hands clasped together in a supportive grip, with some hands having manicured nails. The overall theme is professional collaboration and teamwork.

SCHEDULE **ROLE ASSIGNMENT**

Schedule Role Assignment

ROLE
ASSIGNMENT



7202
Kankrit
Thamwirotsiri



7209
Thanuwach
Thanachoksirirat



7211
Taechapat
Akejiratrakul



7233
Phumin
Boontrarickamas



7236
Punchaya
Chancharoen

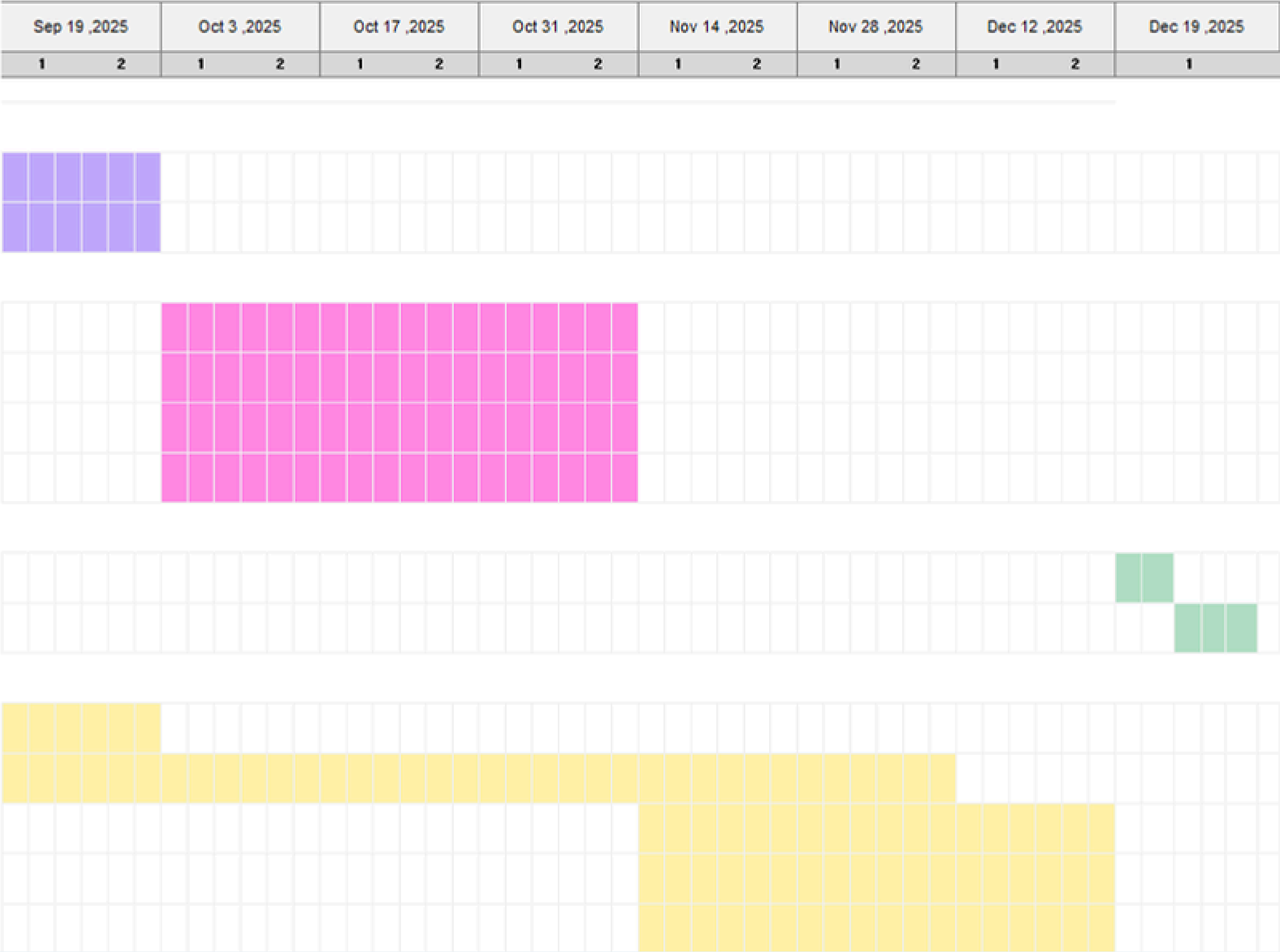
ID	TASK	ASSIGNED TO	PROGRESS	START	END
1	Conceptual Design				
	UseCase Diagram	7202	100%	7/9/25	19/09/2025
	Activity Diagram	7202	100%	7/9/25	19/09/2025
2	System Design				
	Data management	7202, 7209, 7211, 7233, 7236	100%	19/09/2025	31/10/2025
	Specification Design	7233	100%	19/09/2025	31/10/2025
	Structural Design	7236	100%	19/09/2025	31/10/2025
	Database Design	7233	100%	19/09/2025	31/10/2025

3	Test				
	Unit test	7209	100%	12/12/25	14/12/2025
	Operation test	7211	100%	14/12/2025	18/12/2025
4	Data Analyze				
	EDA	7202, 7209, 7211, 7233, 7236	100%	7/9/25	19/09/2025
	Preprocess Data	7233, 7236	100%	19/09/2025	28/11/2025
	Model	7233, 7236	100%	14/11/2025	12/12/25
	Hyparameter Tuning	7233, 7236	100%	14/11/2025	12/12/25
	Evaluate	7233, 7236	100%	14/11/2025	12/12/25

Schedule Role Assignment

SCHEDULE

ID	TASK	ASSIGNED TO	PROGRESS	START	END
1	Conceptual Design				
	UseCase Diagram	7202	100%	7/9/25	19/09/2025
	Activity Diagram	7202	100%	7/9/25	19/09/2025
2	System Design				
	Data management	7202, 7209, 7211, 7233, 7236	100%	19/09/2025	31/10/2025
	Specification Design	7233	100%	19/09/2025	31/10/2025
	Structural Design	7236	100%	19/09/2025	31/10/2025
	Database Design	7233	100%	19/09/2025	31/10/2025
3	Test				
	Unit test	7209	100%	12/12/25	14/12/2025
	Operation test	7211	100%	14/12/2025	18/12/2025
4	Data Analyze				
	EDA	7202, 7209, 7211, 7233, 7236	100%	7/9/25	19/09/2025
	Preprocess Data	7233, 7236	100%	19/09/2025	28/11/2025
	Model	7233, 7236	100%	14/11/2025	12/12/25
	Hyperparameter Tuning	7233, 7236	100%	14/11/2025	12/12/25
	Evaluate	7233, 7236	100%	14/11/2025	12/12/25



OUR TEAM



1014

**Kankrit
Thamwirotsiri**



7209

**Thanuwach
Thanachoksirirat**



7211

**Taechapat
Akejiratrakul**



7233

**Phumin
Boontrarickamas**



7236

**Punchaya
Chancharoen**



DEMO WEBSITE FOR PC



[LINK HERE](#)



Appendix

DIABETES RISK LEVEL

< 30%
Low Risk

HbA1c < 5.7%
BMI < 23

30 – 70%
Medium Risk

HbA1c : 5.7% – 6.4%
BMI > 23

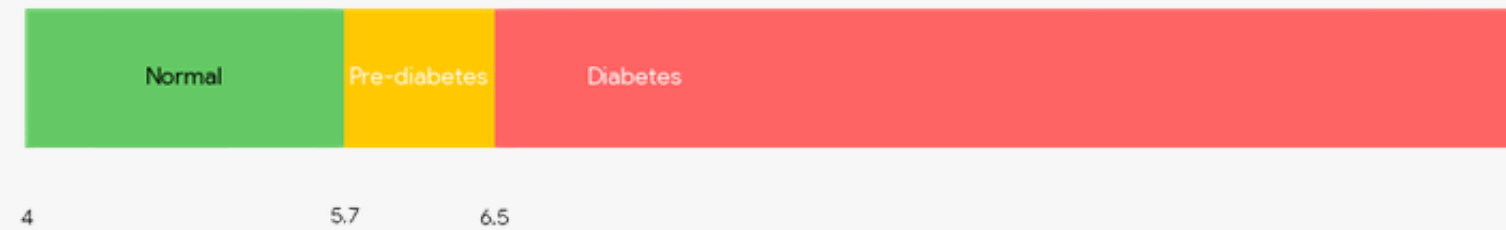
> 70%
High Risk

HbA1c > 6.5%
Fasting Blood Sugar =126 mg/dL

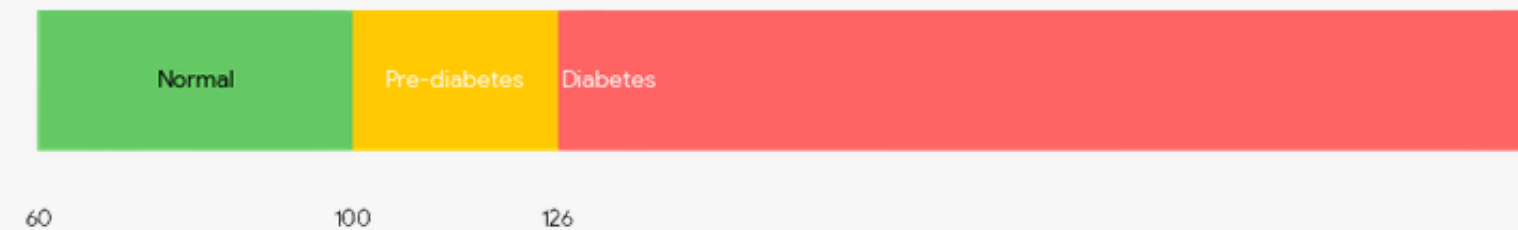
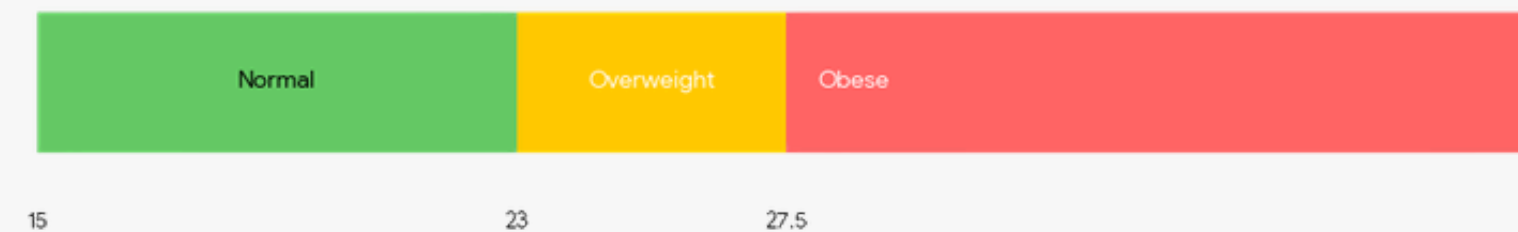
Appendix

DIABETES RISK LEVEL

HbA1c (Hemoglobin A1c) - %



Fasting Blood Sugar (mg/dL)

BMI (Body Mass Index) - kg/m² (Asian Standard)

HbA1c (Hemoglobin A1c)

- **Normal:** Below 5.7%
- **Pre-diabetes:** 5.7% – 6.4%
- **Diabetes:** 6.5% or higher

Fasting Blood Sugar (FBS)

- **Normal:** 70 – 99 mg/dL
- **Pre-diabetes:** 100 – 125 mg/dL
- **Diabetes:** 126 mg/dL or higher

BMI (Body Mass Index) – Asian Criteria

- **Normal:** 18.5 – 22.9 kg/m²
- **Overweight:** 23.0 – 24.9 kg/m²
- **Obese:** 25.0 kg/m² or higher

Appendix

FUNCTION

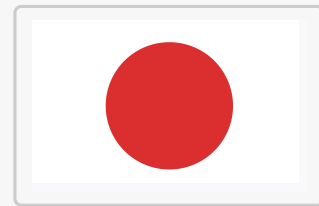
Diabetes Risk Level

0.00 – 0.29	➡	Low
0.30 – 0.69	➡	Medium
0.70 – 1.00	➡	High

BMI Level

< 18.5	➡	Underweight
18.5 – 24.9	➡	Normal
25 – 29.9	➡	Overweight
30 +	➡	Obese

Clean Text



Diabetes
mellitus



Diabetes

List
Columns



Single
Columns

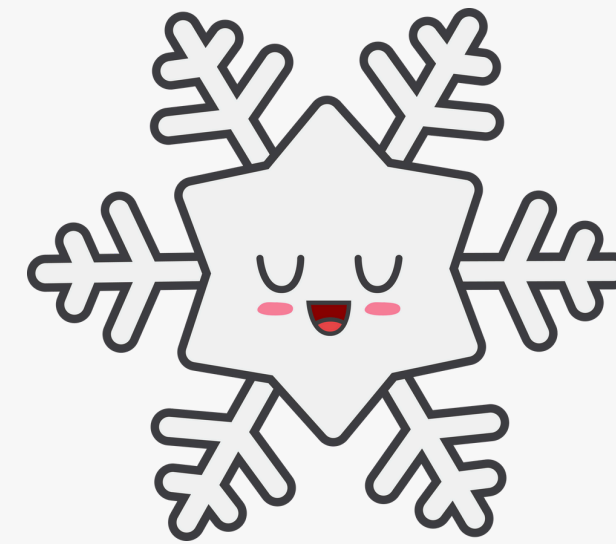
Complex
text



Simple
text

Activity Level

Stay room	➡	Low
Join / Hobby	➡	Medium
Hard work	➡	High



Appendix

CORRERATION'S MODEL

